Problem Statement

Build and train a Convolutional Neural Network (CNN) to classify images of plant leaves into 38 different classes representing various healthy and diseased plant conditions (using the PlantVillage dataset). The workflow involves extracting and exploring the dataset, splitting it into train/validation/test sets, applying data augmentation, training a CNN classifier, evaluating its performance using accuracy, a classification report, and a confusion matrix, and finally testing the trained model on a new, unseen leaf image.

```
# Install the required Python packages: TensorFlow (deep learning),
Pandas (data handling),
# Matplotlib & Seaborn (visualization), scikit-learn (evaluation
metrics), and Pillow (image processing)
!pip install tensorflow pandas matplotlib seaborn scikit-learn pillow

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```

```
# Check whether the zipped dataset file (val.zip) exists at the given
path
import os

zip_path = "/val.zip"

print(os.path.exists(zip_path))
```

```
True
```

```python
# Extract the contents of the zipped dataset into the target directory
import zipfile
import os

zip_path = "/val.zip"
extract_path = "/content/plantvillage"

with zipfile.ZipFile(zip_path, "r") as zip_ref:
    zip_ref.extractall(extract_path)

print("Dataset extracted successfully!")
```

```
Dataset extracted successfully!
```

```python
# List the folders/files inside the extracted dataset directory to
inspect its structure
import os

print(os.listdir("/content/plantvillage"))
```

```
['val']
```

```python
# Set the path to the folder containing the actual class-wise image
data
dataset_path = "/content/plantvillage/val"

# List all class (plant disease) folder names found in the dataset and
print how many there are
import os

classes = sorted(os.listdir(dataset_path))

print("Number of classes:", len(classes))

for i, class_name in enumerate(classes):
    print(i, class_name)
```

```
Number of classes: 38
0 Apple___Apple_scab
1 Apple___Black_rot
2 Apple___Cedar_apple_rust
3 Apple___healthy
4 Blueberry___healthy
5 Cherry_(including_sour)___Powdery_mildew
6 Cherry_(including_sour)___healthy
7 Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot
8 Corn_(maize)___Common_rust_
9 Corn_(maize)___Northern_Leaf_Blight
10 Corn_(maize)___healthy
11 Grape___Black_rot
12 Grape___Esca_(Black_Measles)
```

```
13 Grape___Leaf_blight_(Isariopsis_Leaf_Spot)
14 Grape___healthy
15 Orange___Haunglongbing_(Citrus_greening)
16 Peach___Bacterial_spot
17 Peach___healthy
18 Pepper,_bell___Bacterial_spot
19 Pepper,_bell___healthy
20 Potato___Early_blight
21 Potato___Late_blight
22 Potato___healthy
23 Raspberry___healthy
24 Soybean___healthy
25 Squash___Powdery_mildew
26 Strawberry___Leaf_scorch
27 Strawberry___healthy
28 Tomato___Bacterial_spot
29 Tomato___Early_blight
30 Tomato___Late_blight
31 Tomato___Leaf_Mold
32 Tomato___Septoria_leaf_spot
33 Tomato___Spider_mites Two-spotted_spider_mite
34 Tomato___Target_Spot
35 Tomato___Tomato_Yellow_Leaf_Curl_Virus
36 Tomato___Tomato_mosaic_virus
37 Tomato___healthy
```

```python
# Count and print the number of images present in each class folder,
and sum them up
# to get the total number of images in the dataset
total_images = 0

for class_name in classes:
    class_path = os.path.join(dataset_path, class_name)

    if os.path.isdir(class_path):
        count = len([
            f for f in os.listdir(class_path)
            if f.lower().endswith((".jpg", ".jpeg", ".png"))
        ])

        print(f"{class_name}: {count}")
        total_images += count

print("\nTotal images:", total_images)
```

```
Apple___Apple_scab: 126
Apple___Black_rot: 124
Apple___Cedar_apple_rust: 55
Apple___healthy: 329
Blueberry___healthy: 300
```

```
Cherry_(including_sour)___Powdery_mildew: 210
Cherry_(including_sour)___healthy: 170
Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot: 102
Corn_(maize)___Common_rust_: 238
Corn_(maize)___Northern_Leaf_Blight: 197
Corn_(maize)___healthy: 232
Grape___Black_rot: 236
Grape___Esca_(Black_Measles): 276
Grape___Leaf_blight_(Isariopsis_Leaf_Spot): 215
Grape___healthy: 84
Orange___Haunglongbing_(Citrus_greening): 1101
Peach___Bacterial_spot: 459
Peach___healthy: 72
Pepper,_bell___Bacterial_spot: 199
Pepper,_bell___healthy: 295
Potato___Early_blight: 200
Potato___Late_blight: 200
Potato___healthy: 30
Raspberry___healthy: 74
Soybean___healthy: 1018
Squash___Powdery_mildew: 367
Strawberry___Leaf_scorch: 221
Strawberry___healthy: 91
Tomato___Bacterial_spot: 425
Tomato___Early_blight: 200
Tomato___Late_blight: 381
Tomato___Leaf_Mold: 190
Tomato___Septoria_leaf_spot: 354
Tomato___Spider_mites Two-spotted_spider_mite: 335
Tomato___Target_Spot: 280
Tomato___Tomato_Yellow_Leaf_Curl_Virus: 1071
Tomato___Tomato_mosaic_virus: 74
Tomato___healthy: 318

Total images: 10849
```

```python
# Split the dataset into train (70%), validation (15%), and test (15%)
sets:
# for each class, shuffle its images, divide them according to the
split ratios,
# and copy them into a new directory structure organized by split and
class
import os
import shutil
import random

# Original dataset
source_dir = "/content/plantvillage/val"

# New dataset location
```

```
output_dir = "/content/plant_dataset"

# Split ratios
train_ratio = 0.70
validation_ratio = 0.15
test_ratio = 0.15

# Reproducibility
random.seed(42)

# Create train/validation/test folders
for split in ["train", "validation", "test"]:
    os.makedirs(os.path.join(output_dir, split), exist_ok=True)

# Process each class
for class_name in os.listdir(source_dir):

    class_source = os.path.join(source_dir, class_name)

    # Ignore files; only process directories
    if not os.path.isdir(class_source):
        continue

    # Get image files
    images = [
        f for f in os.listdir(class_source)
        if f.lower().endswith((".jpg", ".jpeg", ".png"))
    ]

    # Shuffle images
    random.shuffle(images)

    # Calculate split points
    total = len(images)

    train_end = int(total * train_ratio)
    validation_end = train_end + int(total * validation_ratio)

    train_images = images[:train_end]
    validation_images = images[train_end:validation_end]
    test_images = images[validation_end:]

    # Create class folders
    for split in ["train", "validation", "test"]:
        os.makedirs(
            os.path.join(output_dir, split, class_name),
            exist_ok=True
        )

    # Copy images
```

```
    for image in train_images:
        shutil.copy2(
            os.path.join(class_source, image),
            os.path.join(output_dir, "train", class_name, image)
        )

    for image in validation_images:
        shutil.copy2(
            os.path.join(class_source, image),
            os.path.join(output_dir, "validation", class_name, image)
        )

    for image in test_images:
        shutil.copy2(
            os.path.join(class_source, image),
            os.path.join(output_dir, "test", class_name, image)
        )

print("Dataset splitting completed successfully!")
```

```
Dataset splitting completed successfully!
```

```
# Verify the split by counting the total number of images in each of
the
# train/validation/test folders
import os

for split in ["train", "validation", "test"]:

    split_path = os.path.join(
        "/content/plant_dataset",
        split
    )

    total = 0

    for class_name in os.listdir(split_path):

        class_path = os.path.join(
            split_path,
            class_name
        )

        if os.path.isdir(class_path):
            count = len([
                f for f in os.listdir(class_path)
                if f.lower().endswith((".jpg", ".jpeg", ".png"))
            ])

            total += count
```

```
    print(split, "images:", total)
```

```
train images: 7577
validation images: 1612
test images: 1660
```

```
# List the class folders present in the newly created training set to
confirm
# all classes were carried over correctly
train_classes = os.listdir("/content/plant_dataset/train")

print("Number of classes:", len(train_classes))

for class_name in sorted(train_classes):
    print(class_name)
```

```
Number of classes: 38
Apple___Apple_scab
Apple___Black_rot
Apple___Cedar_apple_rust
Apple___healthy
Blueberry___healthy
Cherry_(including_sour)___Powdery_mildew
Cherry_(including_sour)___healthy
Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot
Corn_(maize)___Common_rust_
Corn_(maize)___Northern_Leaf_Blight
Corn_(maize)___healthy
Grape___Black_rot
Grape___Esca_(Black_Measles)
Grape___Leaf_blight_(Isariopsis_Leaf_Spot)
Grape___healthy
Orange___Haunglongbing_(Citrus_greening)
Peach___Bacterial_spot
Peach___healthy
Pepper,_bell___Bacterial_spot
Pepper,_bell___healthy
Potato___Early_blight
Potato___Late_blight
Potato___healthy
Raspberry___healthy
Soybean___healthy
Squash___Powdery_mildew
Strawberry___Leaf_scorch
Strawberry___healthy
Tomato___Bacterial_spot
Tomato___Early_blight
Tomato___Late_blight
Tomato___Leaf_Mold
```

```
Tomato___Septoria_leaf_spot
Tomato___Spider_mites Two-spotted_spider_mite
Tomato___Target_Spot
Tomato___Tomato_Yellow_Leaf_Curl_Virus
Tomato___Tomato_mosaic_virus
Tomato___healthy

# Set up image data generators for training, validation, and test sets:
# apply rescaling and augmentation (rotation, shifts, zoom, horizontal flip) to the
# training images to improve generalization, and only rescaling to validation/test
# images. Then load images in batches directly from their respective directories,
# resizing them to 128x128 and one-hot encoding their class labels
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Image dimensions
IMG_SIZE = (128, 128)

# Number of images processed at once
BATCH_SIZE = 32

# Dataset paths
train_path = "/content/plant_dataset/train"
validation_path = "/content/plant_dataset/validation"
test_path = "/content/plant_dataset/test"

# Training data augmentation
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    width_shift_range=0.1,
    height_shift_range=0.1,
    zoom_range=0.2,
    horizontal_flip=True
)

# Validation and test data: only normalization
validation_datagen = ImageDataGenerator(
    rescale=1./255
)

test_datagen = ImageDataGenerator(
    rescale=1./255
)

# Load training images
```

```
train_generator = train_datagen.flow_from_directory(
    train_path,
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode="categorical"
)

# Load validation images
validation_generator = validation_datagen.flow_from_directory(
    validation_path,
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode="categorical"
)

# Load test images
test_generator = test_datagen.flow_from_directory(
    test_path,
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode="categorical",
    shuffle=False
)

print("\nNumber of classes:", train_generator.num_classes)
```

```
Found 7577 images belonging to 38 classes.
Found 1612 images belonging to 38 classes.
Found 1660 images belonging to 38 classes.

Number of classes: 38
```

```
# Print the number of images found in each split and the total number
of classes
print("Training images:", train_generator.samples)
print("Validation images:", validation_generator.samples)
print("Test images:", test_generator.samples)
print("Number of classes:", train_generator.num_classes)
```

```
Training images: 7577
Validation images: 1612
Test images: 1660
Number of classes: 38
```

```
# Print the mapping between class names and their numeric index
assigned by the generator
class_indices = train_generator.class_indices

for class_name, index in class_indices.items():
    print(index, "->", class_name)
```

```
0 -> Apple___Apple_scab
1 -> Apple___Black_rot
2 -> Apple___Cedar_apple_rust
3 -> Apple___healthy
4 -> Blueberry___healthy
5 -> Cherry_(including_sour)___Powdery_mildew
6 -> Cherry_(including_sour)___healthy
7 -> Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot
8 -> Corn_(maize)___Common_rust_
9 -> Corn_(maize)___Northern_Leaf_Blight
10 -> Corn_(maize)___healthy
11 -> Grape___Black_rot
12 -> Grape___Esca_(Black_Measles)
13 -> Grape___Leaf_blight_(Isariopsis_Leaf_Spot)
14 -> Grape___healthy
15 -> Orange___Haunglongbing_(Citrus_greening)
16 -> Peach___Bacterial_spot
17 -> Peach___healthy
18 -> Pepper,_bell___Bacterial_spot
19 -> Pepper,_bell___healthy
20 -> Potato___Early_blight
21 -> Potato___Late_blight
22 -> Potato___healthy
23 -> Raspberry___healthy
24 -> Soybean___healthy
25 -> Squash___Powdery_mildew
26 -> Strawberry___Leaf_scorch
27 -> Strawberry___healthy
28 -> Tomato___Bacterial_spot
29 -> Tomato___Early_blight
30 -> Tomato___Late_blight
31 -> Tomato___Leaf_Mold
32 -> Tomato___Septoria_leaf_spot
33 -> Tomato___Spider_mites Two-spotted_spider_mite
34 -> Tomato___Target_Spot
35 -> Tomato___Tomato_Yellow_Leaf_Curl_Virus
36 -> Tomato___Tomato_mosaic_virus
37 -> Tomato___healthy
```

```
# Build a Convolutional Neural Network (CNN) using the Keras
Sequential API:
# three blocks of Conv2D + MaxPooling2D layers (with increasing filter
sizes: 32, 64, 128)
# to extract spatial features, followed by a Flatten layer, a Dense
hidden layer with
# Dropout for regularization, and a final Dense output layer with
softmax activation
# for classifying images into 38 plant disease classes. Print the
model architecture summary
import tensorflow as tf
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Conv2D,
    MaxPooling2D,
    Flatten,
    Dense,
    Dropout
)

# Create CNN model
model = Sequential([

    # Convolutional Layer 1
    Conv2D(
        32,
        (3, 3),
        activation="relu",
        input_shape=(128, 128, 3)
    ),

    # Pooling Layer 1
    MaxPooling2D(pool_size=(2, 2)),

    # Convolutional Layer 2
    Conv2D(
        64,
        (3, 3),
        activation="relu"
    ),

    # Pooling Layer 2
    MaxPooling2D(pool_size=(2, 2)),

    # Convolutional Layer 3
    Conv2D(
        128,
        (3, 3),
        activation="relu"
    ),

    # Pooling Layer 3
    MaxPooling2D(pool_size=(2, 2)),

    # Convert feature maps to 1D
    Flatten(),

    # Fully connected layer
    Dense(
        128,
        activation="relu"
```

```
    ),

    # Dropout to reduce overfitting
    Dropout(0.5),

    # Output layer
    Dense(
        38,
        activation="softmax"
    )
])

model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/
convolutional/base_conv.py:113: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)
```

```
Model: "sequential"
```

```
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┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━┩
│ conv2d (Conv2D)                      │ (None, 126, 126, 32)        │
896 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ max_pooling2d (MaxPooling2D)         │ (None, 63, 63, 32)          │
0 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ conv2d_1 (Conv2D)                    │ (None, 61, 61, 64)          │
18,496 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ max_pooling2d_1 (MaxPooling2D)       │ (None, 30, 30, 64)          │
0 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ conv2d_2 (Conv2D)                    │ (None, 28, 28, 128)         │
73,856 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
```

```
| max_pooling2d_2 (MaxPooling2D)  | (None, 14, 14, 128)    |
0 |
├─────────────────────────────────┼────────────────────────┼──────────
──────┤
| flatten (Flatten)               | (None, 25088)          |
0 |
├─────────────────────────────────┼────────────────────────┼──────────
──────┤
| dense (Dense)                   | (None, 128)            |
3,211,392 |
├─────────────────────────────────┼────────────────────────┼──────────
──────┤
| dropout (Dropout)               | (None, 128)            |
0 |
├─────────────────────────────────┼────────────────────────┼──────────
──────┤
| dense_1 (Dense)                 | (None, 38)             |
4,902 |
└─────────────────────────────────┴────────────────────────┴──────────
──────┘

 Total params: 3,309,542 (12.62 MB)

 Trainable params: 3,309,542 (12.62 MB)

 Non-trainable params: 0 (0.00 B)
```

```
# Compile the model with the Adam optimizer, categorical crossentropy
loss
# (suitable for one-hot encoded multi-class labels), and accuracy as
the evaluation metric
model.compile(
    optimizer="adam",
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)

print("Model compiled successfully!")
```

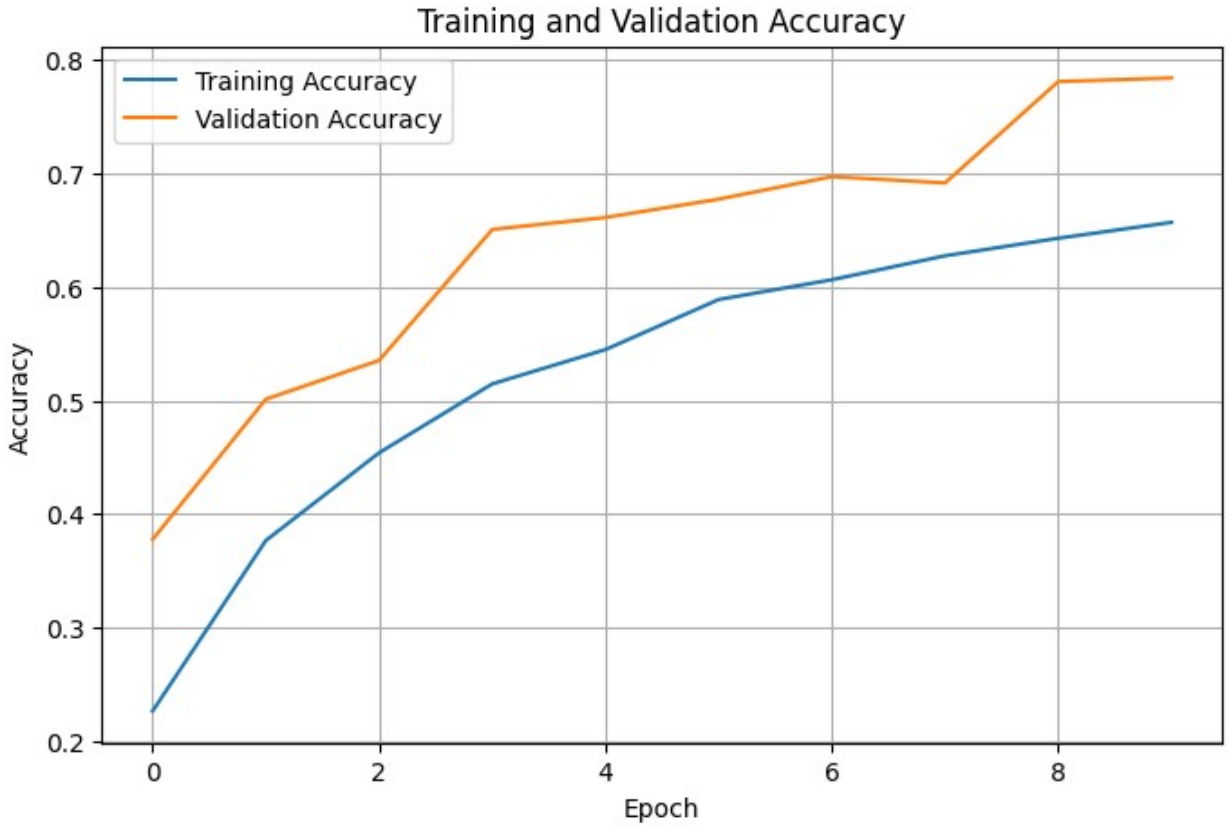
```
Model compiled successfully!
```

```
# Train the CNN on the training data for 10 epochs, validating
performance on
# the validation set after each epoch
history = model.fit(
    train_generator,
    epochs=10,
    validation_data=validation_generator
)
```

```
Epoch 1/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 55s 207ms/step - accuracy: 0.2266 - loss:
2.9936 - val_accuracy: 0.3778 - val_loss: 2.3353
Epoch 2/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 43s 179ms/step - accuracy: 0.3768 - loss:
2.3026 - val_accuracy: 0.5012 - val_loss: 1.8153
Epoch 3/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 43s 180ms/step - accuracy: 0.4540 - loss:
1.9644 - val_accuracy: 0.5354 - val_loss: 1.6402
Epoch 4/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 42s 175ms/step - accuracy: 0.5147 - loss:
1.7125 - val_accuracy: 0.6507 - val_loss: 1.1577
Epoch 5/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 40s 167ms/step - accuracy: 0.5449 - loss:
1.5649 - val_accuracy: 0.6613 - val_loss: 1.1064
Epoch 6/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 40s 168ms/step - accuracy: 0.5890 - loss:
1.4033 - val_accuracy: 0.6774 - val_loss: 1.0661
Epoch 7/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 39s 164ms/step - accuracy: 0.6064 - loss:
1.3427 - val_accuracy: 0.6973 - val_loss: 0.9848
Epoch 8/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 39s 166ms/step - accuracy: 0.6276 - loss:
1.2479 - val_accuracy: 0.6917 - val_loss: 0.9648
Epoch 9/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 39s 164ms/step - accuracy: 0.6430 - loss:
1.1712 - val_accuracy: 0.7810 - val_loss: 0.7204
Epoch 10/10
237/237 ━━━━━━━━━━━━━━━━━━━━ 41s 172ms/step - accuracy: 0.6570 - loss:
1.1329 - val_accuracy: 0.7841 - val_loss: 0.7043
```

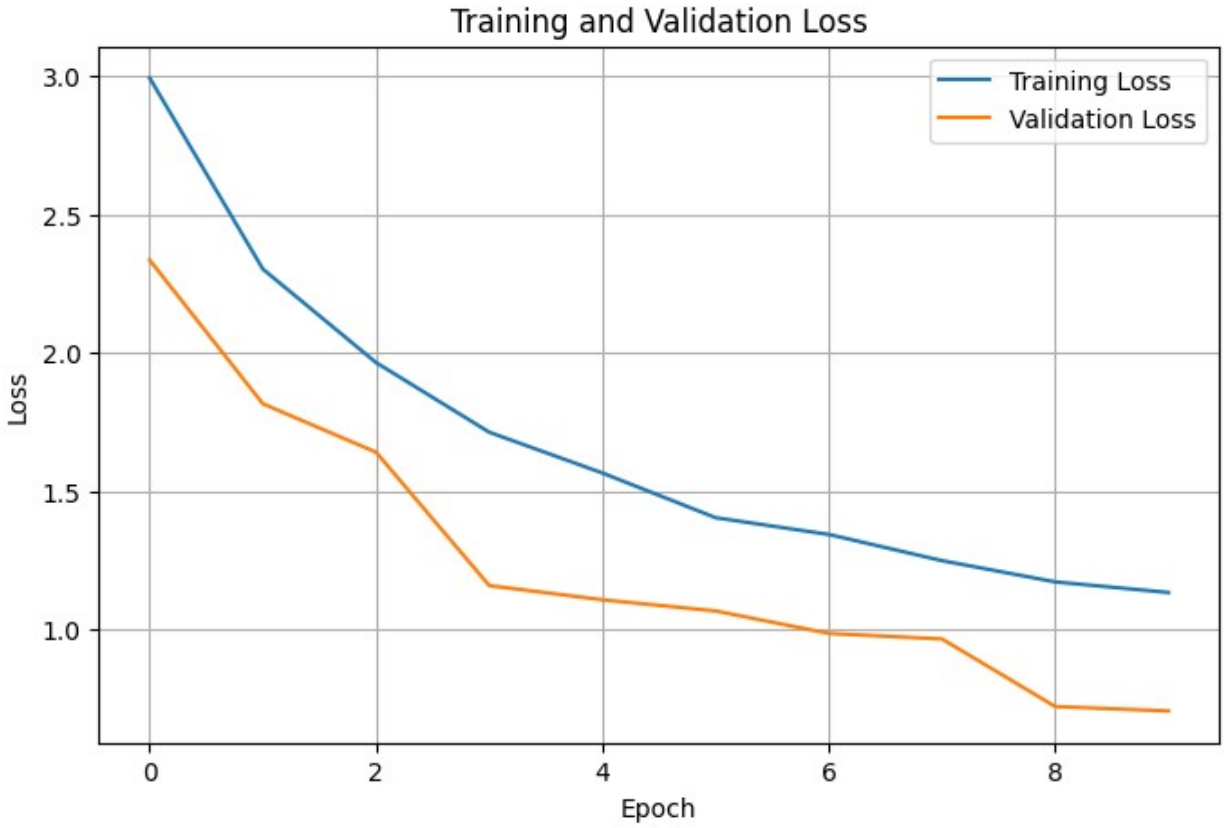
```python
# Plot the training and validation accuracy across epochs to visualize
how well
# the model learned and whether it overfit
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))

plt.plot(
    history.history["accuracy"],
    label="Training Accuracy"
)

plt.plot(
    history.history["val_accuracy"],
    label="Validation Accuracy"
)

plt.xlabel("Epoch")
plt.ylabel("Accuracy")
```

```
plt.title("Training and Validation Accuracy")

plt.legend()
plt.grid(True)

plt.show()
```




```
# Plot the training and validation loss across epochs to visualize the
model's
# convergence and check for overfitting
plt.figure(figsize=(8, 5))

plt.plot(
    history.history["loss"],
    label="Training Loss"
)

plt.plot(
    history.history["val_loss"],
    label="Validation Loss"
)

plt.xlabel("Epoch")
plt.ylabel("Loss")
```

```
plt.title("Training and Validation Loss")

plt.legend()
plt.grid(True)

plt.show()
```




```
# Evaluate the trained model on the unseen test set and print the
resulting
# test loss and test accuracy
test_loss, test_accuracy = model.evaluate(test_generator)

print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy)
```

```
52/52 ━━━━━━━━━━━━━━━━━━━━ 3s 50ms/step - accuracy: 0.7741 - loss:
0.7227
Test Loss: 0.7227166295051575
Test Accuracy: 0.7740963697433472
```

```
# Generate predictions for all test images, convert the predicted
probabilities
# into class labels, and retrieve the true labels and class names for
comparison
```

```python
import numpy as np

predictions = model.predict(test_generator)

predicted_classes = np.argmax(
    predictions,
    axis=1
)

true_classes = test_generator.classes

class_names = list(
    test_generator.class_indices.keys()
)

print("Predictions completed!")
print("Number of predictions:", len(predicted_classes))
```

```
52/52 ━━━━━━━━━━━━━━━━━━━━ 3s 41ms/step
Predictions completed!
Number of predictions: 1660
```

```python
# Print a detailed classification report (precision, recall, F1-score per class)
# comparing the true and predicted labels
from sklearn.metrics import classification_report

report = classification_report(
    true_classes,
    predicted_classes,
    target_names=class_names
)

print(report)
```

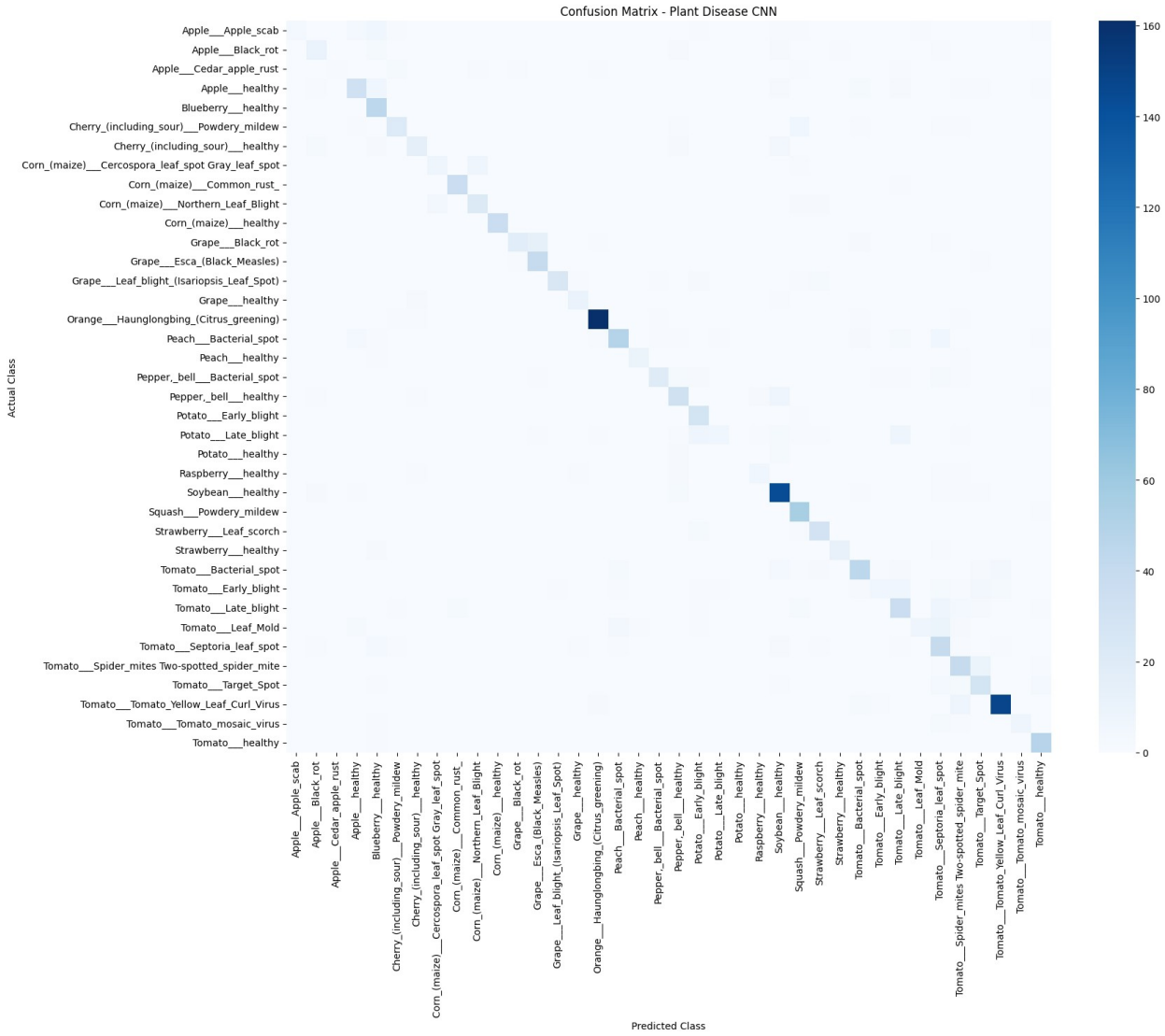
```
                                                    precision
recall  f1-score   support

                                Apple___Apple_scab       1.00
0.20      0.33        20
                                 Apple___Black_rot       0.58
0.55      0.56        20
                             Apple___Cedar_apple_rust       1.00
0.11      0.20         9
                                   Apple___healthy       0.72
0.66      0.69        50
                               Blueberry___healthy       0.62
1.00      0.76        45
          Cherry_(including_sour)___Powdery_mildew       0.75
0.66      0.70        32
                 Cherry_(including_sour)___healthy       0.75
```

```
0.56      0.64        27
Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot       0.62
0.50      0.55        16
                     Corn_(maize)___Common_rust_       0.92
0.97      0.95        37
             Corn_(maize)___Northern_Leaf_Blight       0.73
0.71      0.72        31
                       Corn_(maize)___healthy       1.00
1.00      1.00        36
                            Grape___Black_rot       0.90
0.50      0.64        36
                     Grape___Esca_(Black_Measles)       0.71
0.95      0.82        42
       Grape___Leaf_blight_(Isariopsis_Leaf_Spot)       0.96
0.76      0.85        33
                         Grape___healthy       0.79
0.79      0.79        14
         Orange___Haunglongbing_(Citrus_greening)       0.96
0.97      0.97       166
                       Peach___Bacterial_spot       0.83
0.70      0.76        70
                         Peach___healthy       0.91
0.83      0.87        12
                    Pepper,_bell___Bacterial_spot       0.86
0.61      0.72        31
                       Pepper,_bell___healthy       0.65
0.71      0.68        45
                        Potato___Early_blight       0.62
0.97      0.75        30
                         Potato___Late_blight       0.75
0.20      0.32        30
                          Potato___healthy       0.00
0.00      0.00         5
                       Raspberry___healthy       0.67
0.50      0.57        12
                         Soybean___healthy       0.82
0.94      0.87       154
                      Squash___Powdery_mildew       0.74
0.98      0.85        56
                     Strawberry___Leaf_scorch       0.77
0.88      0.82        34
                        Strawberry___healthy       0.92
0.80      0.86        15
                      Tomato___Bacterial_spot       0.75
0.69      0.72        65
                        Tomato___Early_blight       0.67
0.20      0.31        30
                         Tomato___Late_blight       0.60
0.62      0.61        58
```

```
                                Tomato___Leaf_Mold       0.88
0.24      0.38        29
                 Tomato___Septoria_leaf_spot       0.48
0.74      0.58        54
     Tomato___Spider_mites Two-spotted_spider_mite       0.59
0.78      0.67        51
                        Tomato___Target_Spot       0.58
0.69      0.63        42
          Tomato___Tomato_Yellow_Leaf_Curl_Virus       0.94
0.91      0.93       162
                 Tomato___Tomato_mosaic_virus       1.00
0.58      0.74        12
                           Tomato___healthy       0.81
0.98      0.89        49

                                    accuracy
0.77      1660
                                   macro avg       0.76
0.67      0.68      1660
                                weighted avg       0.79
0.77      0.76      1660


/usr/local/lib/python3.12/dist-packages/sklearn/metrics/
_classification.py:1565: UndefinedMetricWarning: Precision is ill-
defined and being set to 0.0 in labels with no predicted samples. Use
`zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classificatio
n.py:1565: UndefinedMetricWarning: Precision is ill-defined and being
set to 0.0 in labels with no predicted samples. Use `zero_division`
parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classificatio
n.py:1565: UndefinedMetricWarning: Precision is ill-defined and being
set to 0.0 in labels with no predicted samples. Use `zero_division`
parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
```

```
# Compute the confusion matrix and visualize it as a heatmap to see
how well
# the model distinguishes between the different plant disease classes
from sklearn.metrics import confusion_matrix
import seaborn as sns

cm = confusion_matrix(
    true_classes,
```

```
    predicted_classes
)

plt.figure(figsize=(18, 15))

sns.heatmap(
    cm,
    cmap="Blues",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("Confusion Matrix - Plant Disease CNN")

plt.xticks(rotation=90)
plt.yticks(rotation=0)

plt.tight_layout()
plt.show()
```




```
# Create a directory for saved models (if it doesn't already exist) and save
# the trained CNN model to disk in Keras format
import os

os.makedirs("/content/models", exist_ok=True)

model.save("/content/models/plant_disease_cnn.keras")

print("Model saved successfully!")
```

```
Model saved successfully!
```

```
# Test the trained model on a new, single external image: load and resize it,
# display it, convert it into a normalized array with a batch dimension, run it
```

```python
# through the model, and print the predicted class along with the
confidence score
from tensorflow.keras.preprocessing import image
import numpy as np
import matplotlib.pyplot as plt

# Path to your new image
img_path = "/tomato_leaf.webp"

# Load and resize image
img = image.load_img(
    img_path,
    target_size=(128, 128)
)

# Display image
plt.figure(figsize=(5, 5))
plt.imshow(img)
plt.axis("off")
plt.title("Input Image")
plt.show()

# Convert image to array
img_array = image.img_to_array(img)

# Normalize pixel values
img_array = img_array / 255.0

# Add batch dimension
img_array = np.expand_dims(img_array, axis=0)

# Make prediction
prediction = model.predict(img_array)

# Get predicted class
predicted_index = np.argmax(prediction[0])

predicted_class = class_names[predicted_index]

# Get confidence
confidence = prediction[0][predicted_index] * 100

print("Predicted Class:", predicted_class)
print("Confidence: {:.2f}%".format(confidence))
```

Input Image

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 881ms/step
Predicted Class: Tomato___Septoria_leaf_spot
Confidence: 38.16%
```